

after creating the 2 tables (orders, products)

1- Test: insert a product with a negative price and confirm it is rejected. Insert two products with the same name and confirm the second is rejected

```
Data Output Messages Notifications
ERROR: new row for relation "products" violates check constraint
"products_price_check"
Failing row contains (1, ranch, -500).

SQL state: 23514
Detail: Failing row contains (1, ranch, -500).
```

one rejected

```
Data Output Messages Notifications
INSERT 0 1

Query returned successfully in 51 msec.
```

other with positive price accepted

2- Test: insert an order without a quantity and confirm it defaults to 1.

Data Output

Messages

Notifications

≡+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

Showing rows: 1 to 1

✎

🖨️

Page No:

1

of 1

⏪

⏴

⏵

⏩

|   | <div>order_id</div> <div>[PK] integer ✎</div> | <div>product_id</div> <div>integer ✎</div> | <div>quantity</div> <div>numeric ✎</div> |
|---|-----------------------------------------------|--------------------------------------------|------------------------------------------|
| 1 | 1                                             | 1                                          | 1                                        |

3-part c: Insert two valid products, then two valid orders that reference them. Then attempt to insert an order that references a product\_id that does not exist, and record the error message.

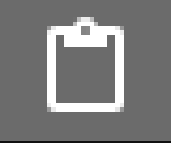
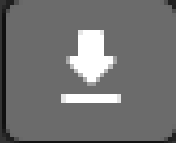
| Data Output                                                                                                                                                                                                                                                                    | Messages | Notifications |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|
| <div> ERROR: insert or update on table "orders" violates foreign key constraint "orders_product_id_fkey"<br/> Key (product_id)=(4) is not present in table "products".<br/><br/> SQL state: 23503<br/> Detail: Key (product_id)=(4) is not present in table "products". </div> |          |               |

4-Part D: Attempt to delete a product that has an existing order. Record the error message, then delete the order first, then the product.

| Data Output                                                                                                                                                                                                                                                                                                  | Messages | Notifications |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|
| <div> ERROR: update or delete on table "products" violates foreign key constraint "orders_product_id_fkey" on table "orders"<br/> Key (product_id)=(1) is still referenced from table "orders".<br/><br/> SQL state: 23503<br/> Detail: Key (product_id)=(1) is still referenced from table "orders". </div> |          |               |

5-Part E: Recreate the orders table so that product\_id uses ON DELETE CASCADE. Insert a

product and an order referencing it, then delete the product and confirm the order disappears automatically

|                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |  |  |  |  |
| Showing rows: 1 to 3                                                              |                                                                                   |                                                                                   |                                                                                   |  |  | Page No:                                                                            |                                                                                     |                                                                                     |
|                                                                                   | product_id<br>[PK] integer                                                        |                                                                                   | name<br>text                                                                      |                                                                                   | price<br>numeric                                                                   |                                                                                     |                                                                                     |                                                                                     |
| 1                                                                                 | 2                                                                                 |                                                                                   | bread                                                                             |                                                                                   | 250                                                                                |                                                                                     |                                                                                     |                                                                                     |
| 2                                                                                 | 3                                                                                 |                                                                                   | butter                                                                            |                                                                                   | 400                                                                                |                                                                                     |                                                                                     |                                                                                     |
| 3                                                                                 | 1                                                                                 |                                                                                   | ranch                                                                             |                                                                                   | 500                                                                                |                                                                                     |                                                                                     |                                                                                     |
|                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |

Data Output

Messages

Notifications

≡+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

SQL

Showing rows: 1 to 1

✎

🖨️

Page No:

1

|   | <div>order_id</div> <div>[PK] integer ✎</div> | <div>product_id</div> <div>integer ✎</div> | <div>quantity</div> <div>numeric ✎</div> |
|---|-----------------------------------------------|--------------------------------------------|------------------------------------------|
| 1 | 1                                             | 1                                          | 1                                        |

| Data Output | Messages | Notifications |
|-------------|----------|---------------|
|-------------|----------|---------------|

DELETE 1

Query returned successfully in 67 msec.

Data Output

Messages

Notifications

≡+

▼

▼

SQL

order\_id

[PK] integer

product\_id

integer

quantity

numeric

order disappeared

6- Practical Challenge:

Showing rows: 1 to 3





Page No:

1

of 1

|   | product_id<br>[PK] integer  | name<br>text  | price<br>numeric  | category_id<br>integer  |
|---|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1 | 1                                                                                                            | ranch                                                                                          | 500                                                                                                  | 1                                                                                                          |
| 2 | 2                                                                                                            | bread                                                                                          | 250                                                                                                  | 2                                                                                                          |
| 3 | 3                                                                                                            | butter                                                                                         | 400                                                                                                  | 3                                                                                                          |

Data Output
Messages
Notifications

ERROR: update or delete on table "categories" violates RESTRIC  
tsetting of foreign key constraint "fk\_product\_category" on table  
"products"  
Key (category\_id)=(1) is referenced from table "products".  
  
SQL state: 23001  
Detail: Key (category\_id)=(1) is referenced from table "products".

final select statements for products, orders, and categories

Showing rows: 1 to 3





Page No:

1

of 1

|   | product_id<br>[PK] integer  | name<br>text  | price<br>numeric  | category_id<br>integer  |  |
|---|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--|
| 1 | 1                                                                                                              | ranch                                                                                            | 500                                                                                                   | 1                                                                                                            |  |
| 2 | 2                                                                                                              | bread                                                                                            | 250                                                                                                   | 2                                                                                                            |  |
| 3 | 3                                                                                                              | butter                                                                                           | 400                                                                                                   | 3                                                                                                            |  |

Showing rows: 1  
to 2



Page  
No:

1

|   | <b>order_id</b><br>[PK] integer | <b>product_id</b><br>integer | <b>quantity</b><br>numeric |
|---|---------------------------------|------------------------------|----------------------------|
| 1 | 2                               | 2                            | 4                          |
| 2 | 3                               | 2                            | 6                          |

Showing rows: 1  
to 3



Page  
No:

1

|   | <b>category_id</b><br>[PK] integer | <b>category_name</b><br>character varying (50) |
|---|------------------------------------|------------------------------------------------|
| 1 | 1                                  | Electronics                                    |
| 2 | 2                                  | Clothing                                       |
| 3 | 3                                  | Food                                           |

Edit range

